ROOT CAUSE UNLOCKED · CONDITION KIT

The Fibromyalgia *Movement* and Pacing Guide

The intervention with the best evidence in this condition, and the one most likely to set you back if the dose is wrong

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

The Fibromyalgia Movement and Pacing Guide

The intervention with the best evidence in this condition, and the one most likely to set you back if the dose is wrong

BEFORE YOU START

Contents

Read this part first

This guide is educational. It is not medical advice, it does not diagnose fibromyalgia or anything else, and reading it does not make you my patient.

Some symptoms need evaluation before an exercise plan, not after one. New weakness, numbness that follows a nerve path, unexplained weight loss, fever, night pain that wakes you and will not settle, a new severe headache, chest pain or shortness of breath on exertion, or any sudden change in what your body does. Widespread pain is common. Those things are not, and they get looked at first.

If you have not had a medical evaluation for widespread pain, get one before you start here. Thyroid disease, anemia, vitamin D deficiency, inflammatory arthritis, sleep apnea and several other conditions produce a pain-and-fatigue picture that looks like fibromyalgia and is treated completely differently. The lab guide in this program covers what to ask for.

One condition-specific caution that has its own section below. If you experience post-exertional malaise, meaning a delayed and disproportionate crash a day or two after activity, the standard advice to increase steadily every week is not appropriate for you and has caused real harm in people with that pattern. Read the screening section in Part One before you start anything.

Why this guide exists

Look at what you have been given for fibromyalgia. There is a guide for your labs, a guide for your food, and a guide for your supplements.

The thing with the best evidence behind it did not have a guide at all.

That is not a small gap. Across the treatment options in this condition, movement is the one that the international guideline puts at the front, and it is the only one where the research is deep enough to argue about *dose* rather than about whether it works (PMID 27377815). Meanwhile the supplement guide in this program runs to thousands of words about compounds with far thinner support, and I say so in that guide too.

So this document is the missing piece. It is also the one I am most careful writing, because generic exercise advice does more damage in fibromyalgia than in any other condition I treat.

What you are actually buying, stated plainly

The evidence here is better than anywhere else in this program, and it is still not a promise.

What is solid: exercise beats no exercise in this condition, across multiple randomized trials and Cochrane reviews. That is about as settled as anything gets here.

What is genuinely uncertain: which *kind* of exercise is best. A 2025 blinded randomized trial compared progressive strength training, constant-intensity strength training, and walking. All three improved symptoms within their own groups. **There were no significant differences between them**, and no measure reached the threshold for a clinically important change (PMID 41674030). Read that honestly and it says two things at once. Doing something beats doing nothing. And arguing about which something is mostly arguing.

What the effect size actually is: modest. In the aerobic exercise dosing analysis, the pooled difference was small (PMID 39805734). Nobody is being cured here. People are getting function back, and function is what I would have you measure.

So I am not going to tell you movement fixes fibromyalgia. I am going to tell you it is the highest-yield thing on the list, that the dose is where people fail, and that this guide is mostly about the dose.

Part one: before you start

The physiology that explains why this feels unfair

Here is something worth knowing before your first session, because it validates an experience you have probably been told is in your head.

In healthy people, a bout of exercise produces short-term pain relief. It has a name, exercise-induced hypoalgesia, and it is one reason exercise feels good afterward.

In fibromyalgia, that response is absent. In a study comparing people with fibromyalgia, people with knee osteoarthritis, and pain-free controls, a five-minute submaximal isometric contraction produced exercise-induced hypoalgesia in the control group and produced **none** in either chronic pain group. Both chronic pain groups also showed lower vagal activity at rest (PMID 36711216). The same inconsistency shows up in a systematic review of isometric exercise across musculoskeletal pain populations, where the effect that appears reliably in healthy people did not appear reliably in people with pain (PMID 33601254).

Sit with what that means. When a healthy person exercises, their nervous system hands them a reward. When you exercise, it often does not. You are being asked to do the same work for none of the immediate payoff.

That is not a character problem and it is not deconditioning. It is a measured difference in how your nervous system responds to exertion.

And here is the part that makes it worth doing anyway. The benefit in fibromyalgia does not come from the session. It comes from the training effect across weeks. Impaired pain modulation has been shown to improve after exercise interventions rather than after single bouts (PMID 35399187, PMID 36660198), and the proposed mechanisms for why exercise helps in this condition are specifically about that longer adaptation (PMID 37484924).

So the deal is honest but harsh: you pay up front and you collect later. Which is exactly why the dose has to be set so the payments are affordable.

The screening question that changes everything

Do you experience post-exertional malaise?

That means a crash that arrives **delayed**, typically twelve to forty-eight hours after activity, that is **disproportionate** to what you did, and that involves a worsening of your whole symptom picture rather than just sore muscles.

Ordinary post-exercise soreness is not this. Soreness arrives within a day, sits in the muscles you used, and fades. Post-exertional malaise can arrive two days later, after a normal grocery trip, and take a week.

If the answer is yes, the standard advice does not apply to you. Programs built on steady weekly increases regardless of how you feel have caused documented harm in people with this pattern, which is why current review of that population treats pacing rather than graded escalation as the appropriate approach (PMID 41366804). The distinction is not academic and it is not a matter of attitude.

If the answer is yes: work from Part Three, the pacing model, and treat every progression rule in Part Two as optional rather than scheduled.

If the answer is no: you can use Part Two's progression as written, still starting lower than you think you should.

If you are not sure, assume yes for the first month and find out. The cost of going too slowly is a slower start. The cost of going too fast is losing weeks.

Three honest expectations before your first session

You will not feel better afterward, at first. See the physiology above. Judge this at six to eight weeks, not at session three.

The first two weeks should feel absurdly easy. If your first session feels like a real workout, it was too much. This is the single most common failure and it is the reason people conclude exercise makes fibromyalgia worse.

Consistency beats intensity by a wide margin. In the dosing analysis, sessions as short as ten minutes appeared in the trials that worked, and useful interventions ran anywhere from three to twenty-four weeks (PMID 39805734). Ten repeatable minutes beats forty minutes you do twice and abandon.

Part two: the four kinds of movement, and what each one has behind it

Aerobic movement, the base of the whole thing

What it is. Anything that raises your breathing and heart rate and that you can keep going: walking, stationary cycling, swimming, pool walking, dancing to music.

The evidence. This is the most studied form in fibromyalgia and has its own Cochrane review (PMID 28636204), sitting inside a broader Cochrane overview of physical activity for chronic pain (PMID 28436583). A dedicated 2025 systematic review analyzed seventeen randomized trials specifically to work out the dose, using the frequency, intensity, type, time, volume and progression framework, and found in favor of aerobic exercise overall (PMID 39805734). An unsupervised walking program has its own randomized trial in women with fibromyalgia (PMID 32544346), which matters because it means this does not require a gym or a supervisor.

The dose, from that analysis. The trials that worked used frequencies from one to ten sessions per week, intensities from light to vigorous, sessions from ten to forty-five minutes, and programs from three to twenty-four weeks. Types included walking, stationary cycling, swimming, pool-based work, interval formats and music-based exercise.

What I take from that range. The successful trials are all over the map, which tells you the specific prescription matters less than finding one you will repeat. So: **start at ten minutes, three times a week, at an intensity where you could hold a conversation.** That is the floor, and the floor is deliberately embarrassing.

Progression, if you do not have post-exertional malaise. Add roughly ten percent to the duration only after two full weeks at the current amount with no increase in your symptom baseline. Duration first. Intensity much later, if at all.

Strength work, which does more than people expect

The evidence. A multi-center assessor-blinded randomized trial put 130 women with fibromyalgia through fifteen weeks of progressive resistance exercise, twice a week, against an active control. The resistance group improved significantly on knee extension strength, overall health status, current pain intensity, six-minute walk distance, elbow flexion strength, pain disability and pain acceptance (PMID 26084281).

Notice what is in that list. Not just strength. **Pain intensity, pain disability, and pain acceptance.** Getting stronger changed how much the pain cost, not only what the muscles could do.

One design detail from that trial is worth copying. The researchers used a person-centred model built specifically to support the participants' confidence about exercising, and they said plainly why: because activity-induced pain is a known risk in this condition. They designed around the flare rather than pretending it would not happen. That is the right posture and it is the one I use.

Stretching alone is not the same thing. A randomized trial comparing muscle stretching against resistance training found they are not interchangeable (PMID 29185675). Stretching has a place. It is not the strength program.

The dose. Twice a week. Start with body weight or the lightest band you own, one set of eight to ten repetitions per movement, five or six movements covering legs, pushing, pulling and trunk. Add repetitions before you add load. Take the first month to build the habit, not the strength.

And the honest counterweight. That 2025 blinded trial found progressive intensity did not beat constant moderate intensity or walking (PMID 41674030). So progress if it feels available. Do not treat progression as the point.

Water, which is where I start most people who are struggling

Why it is different. Buoyancy removes the load while keeping the movement, warmth helps the muscular guarding, and the water gives resistance in every direction without any equipment.

The evidence. Aquatic exercise in fibromyalgia has its own systematic review (PMID 38540665) plus a separate systematic review finding improvement specifically in self-reported sleep quality (PMID 37847348), which matters in a condition where sleep and pain feed each other. Pool-based work also appears among the effective aerobic formats in the dosing analysis (PMID 39805734), and comparative work on which therapeutic exercises help pain most in women with fibromyalgia includes aquatic modalities (PMID 40319533).

Who I put in the water first. Anyone whose land-based attempts have all ended in flares, anyone with significant joint pain layered on top of the fibromyalgia, and anyone who has become fearful of movement. Warm water lowers the stakes of the first attempt, and the first attempt is the one that decides whether there is a second.

Mind-body movement, and the trial that surprised people

The finding. A 52-week single-blind comparative effectiveness randomized trial in 226 adults with fibromyalgia compared classic Yang style tai chi against supervised aerobic exercise, the standard core treatment. Everyone improved. **The tai chi groups improved more** on the fibromyalgia impact questionnaire at 24 weeks, and the trial was published in the BMJ (PMID 29563100).

That is not a small alternative-medicine study. It is a large, long, properly designed trial in a major journal, and it found the gentle option beat the conventional one.

Why I think that is, though this part is my reasoning rather than the trial's finding. Tai chi combines low-intensity movement, balance work, breathing and attention, at a dose almost nobody flares on. In a condition where the main obstacle is that people cannot tolerate the dose, an intervention that is easy to tolerate has an enormous advantage that has nothing to do with the movements themselves.

What this changes practically. If you have failed at conventional exercise repeatedly, tai chi or qigong is not a consolation prize. It is the option with a head-to-head win.

What about education alongside the movement

Pain neuroscience education combined with resistance training has been studied in women with fibromyalgia (PMID 40686548), and low-intensity exercise has been shown to improve pain catastrophizing and other psychological measures (PMID 32455853). I mention this because understanding why your nervous system does what it does is not separate from the physical work. It changes how you interpret a hard day, and how you interpret a hard day determines whether you go back on Thursday.

Part three: pacing, which is the actual skill

Everything above is the menu. This is the method, and if you only take one thing from this guide, take this.

Pace by what is repeatable, not by what you can manage

The instinct is to set the dose by capacity: do what you can today. In fibromyalgia that produces the sawtooth. Good day, do everything, lose the week, recover, repeat, and after a year you are exactly where you started and you have concluded that exercise does not work for you.

The rule that breaks the cycle: find the amount you could do on a bad day, and do that amount on every day, including the good ones.

This is genuinely hard, and it is hard in a specific way. It requires you to stop on a good day while you still feel fine. That feels like leaving value on the table. It is not. It is the only way the baseline itself moves.

Pacing is a real technique with real research, including its complications

Activity pacing has been studied directly, and the literature is more interesting than the slogan version. One study found pacing associated with **both better and worse** symptoms depending on how it was being used (PMID 27322396), and related work on activity management patterns found that the pattern matters more than the label (PMID 28591081). Patients themselves have been asked what they think of pacing and of structured pacing frameworks (PMID 26385155).

The distinction those studies point at is this. **Pacing used to expand what you do is helpful. Pacing used to justify doing less and less is not.** The same word covers both, which is why the direction of travel matters more than the technique.

So the test I would apply monthly: is my baseline going up, holding, or quietly shrinking. If it is shrinking, this has stopped being pacing.

The practical version

Set a floor. Pick the amount you are confident you could do on a bad day. Halve it. That is week one.

Hold it for two weeks. Every scheduled day, including good ones, including days you feel capable of much more.

Then evaluate on the baseline, not on the sessions. Did your general symptom level get worse, stay the same, or improve. Only if it stayed the same or improved do you increase.

Increase by about ten percent, duration first.

When a flare happens, and it will: drop to half your current amount rather than to zero. Going to zero is what turns a three-day flare into a three-week restart. Hold the half amount until the flare settles, then return to where you were, not to where you were heading.

Build in rest days deliberately rather than taking them when forced.

Part four: how to know it is working

Pain is a poor instrument here. It moves for reasons that have nothing to do with your program, and in a condition this variable, judging by pain alone will mislead you in both directions.

Track function instead. Four things, once a week, thirty seconds:

How far can I walk without paying for it tomorrow. One number, in minutes or blocks.

How many days this week did I do my planned movement. Adherence, not intensity. This is the number that predicts everything else.

How many crash days did I have. Days where the whole picture worsened and normal activity stopped.

One thing I did this week that I could not do last month. Written in words, not numbers. Carried the laundry upstairs. Stood through a whole conversation. Went to the thing and stayed.

The tracker at rootcauseunlocked.com/tracker is free and does this for you. In this condition it earns its place, because the changes are gradual and memory across weeks is unreliable in exactly the direction that makes people quit.

Judge the program at eight weeks, not at two. And judge it on those four numbers, not on how any single session felt.

When to stop and get evaluated instead

- New weakness, numbness in a nerve pattern, or loss of bladder or bowel control. Same day.
- Chest pain, unusual shortness of breath, or fainting with exertion. Same day.
- A flare that has not settled after two weeks at half dose.
- Steadily shrinking capacity over two months despite honest pacing. That pattern deserves a fresh look at the diagnosis rather than a harder program.
- Post-exertional malaise that is worsening rather than stabilizing.
- Any new symptom that does not fit the pattern you know.

Working with Root Health

This guide is general education by design, written so it is useful without knowing anything about your case. Personalized work is different: it starts with your history, your other conditions, what you have already tried and how it went, and it sequences the changes instead of stacking them.

If you want that, a Root Discovery Session is free and is the right first step. Book at rootcauseunlocked.com/discovery.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the symptom tracker at rootcauseunlocked.com/tracker is free and works anywhere.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

This guide is educational. It is not medical advice, diagnosis, or treatment, and reading it does not create a doctor and patient relationship.

References

1. Macfarlane GJ, Kronisch C, Dean LE, et al. EULAR revised recommendations for the management of fibromyalgia. *Ann Rheum Dis*. 2017. PMID 27377815
2. Bidonde J, Busch AJ, Schachter CL, et al. Aerobic exercise training for adults with fibromyalgia. *Cochrane Database Syst Rev*. 2017. PMID 28636204
3. Geneen LJ, Moore RA, Clarke C, et al. Physical activity and exercise for chronic pain in adults: an overview of Cochrane reviews. *Cochrane Database Syst Rev*. 2017. PMID 28436583
4. Casanova-Rodriguez D, Ranchal-Sanchez A, Rodriguez RB, Jurado-Castro JM. Aerobic exercise prescription for pain reduction in fibromyalgia: a systematic review and meta-analysis. *Eur J Pain*. 2025. PMID 39805734 (see published erratum, *Eur J Pain* 2025;29(9):e70105)
5. Wang C, Schmid CH, Fielding RA, et al. Effect of tai chi versus aerobic exercise for fibromyalgia: comparative effectiveness randomized controlled trial. *BMJ*. 2018. PMID 29563100
6. Larsson A, Palstam A, Lofgren M, et al. Resistance exercise improves muscle strength, health status and pain intensity in fibromyalgia: a randomized controlled trial. *Arthritis Res Ther*. 2015. PMID 26084281
7. Pontes-Silva A, Dibai-Filho AV, Melo TS, et al. Strength training versus walking on the fibromyalgia impact: a blinded randomised controlled trial. *Musculoskeletal Care*. 2025. PMID 41674030
8. Aquatic exercise in physical therapy treatment for fibromyalgia: systematic review. *Healthcare (Basel)*. 2024. PMID 38540665
9. Aquatic therapy improves self-reported sleep quality in fibromyalgia patients: a systematic review. *Sleep Breath*. 2024. PMID 37847348
10. Bossenger NR, Lewis GN, Rice DA, Shepherd D. The autonomic and nociceptive response to acute exercise is impaired in people with knee osteoarthritis. *Neurobiol Pain*. 2023. PMID 36711216
11. Bonello C, Girdwood M, De Souza K, et al. Does isometric exercise result in exercise induced hypoalgesia in people with local musculoskeletal pain? A systematic review. *Phys Ther Sport*. 2021. PMID 33601254
12. Impaired conditioned pain modulation was restored after a single exercise session in individuals with and without fibromyalgia. *Pain Rep*. 2022. PMID 35399187
13. The effects of a 15-week physical exercise intervention on pain modulation in fibromyalgia. *Neurobiol Pain*. 2023. PMID 36660198
14. Mechanisms hypothesized for pain-relieving effects of exercise in fibromyalgia: a scoping review. *Ther Adv Musculoskelet Dis*. 2023. PMID 37484924
15. Promoting unsupervised walking in women with fibromyalgia: a randomized controlled trial. *Psychol Health Med*. 2021. PMID 32544346
16. Low-intensity physical exercise improves pain catastrophizing and other psychological and physical aspects in women with fibromyalgia. *Int J Environ Res Public Health*. 2020. PMID 32455853
17. Muscle stretching exercises and resistance training in fibromyalgia: which is better? A three-arm randomized controlled trial. *Eur J Phys Rehabil Med*. 2018. PMID 29185675
18. The most effective therapeutic exercises for pain intensity in women with fibromyalgia: a systematic review with network meta-analysis. *Braz J Phys Ther*. 2025. PMID 40319533

19. Pain neuroscience education and resistance training in women with fibromyalgia. *Pain Res Manag*. 2025. PMID 40686548

20. Activity pacing is associated with better and worse symptoms for patients with long-term conditions. *Clin J Pain*. 2017. PMID 27322396

21. Pain-related activity management patterns and function in patients with fibromyalgia syndrome. *Clin J Pain*. 2018. PMID 28591081

22. Exploring patients' opinions of activity pacing and a new activity pacing questionnaire for chronic pain and fatigue. *Physiotherapy*. 2016. PMID 26385155

23. Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS): diagnosis and management. *J Transl Med*. 2025. PMID 41366804